Introduction to Causal Inference

Workshop MPIAE Retreat 2026, Hofgeismar

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12.08.2026

- Quick introduction round
- Basic causality
- Causal Graphs (DAGs)
- Estimating of causal effects (with R)

Interspersed with short exercises. Bio-break somewhere in the middle.

Please state shortly

- Name, department
- Research interests
- Most urgent causal question in your field not answered yet OR Last causal claim in your work.

Introduction

- The nomothetic sciences aim to discover and quantify laws of nature.
- Compact modeling of "reality" for predictions and practical applications.
- Models are usually formulated as if-then statements (If X, then Y) or as functional relationships
$$Y = f(X).$$
- Causality as a fundamental axiom: Nothing happens without a reason.
- However, the question of "what" precedes the question of "why".
- Two (?) types of causality: causal/temporal vs. structural/functional.

Causality

- Causal reasons imply a time arrow (and vice versa).
- Use the concept of an event.
- An event is an (instantaneous) change of state in or on a system.
- Events have direct or indirect effects and can propagate through causal chains.
- Multi-causality is the rule.

- Structural/functional causes are, so to speak, “timeless”.
- Functional changes in a variable caused by other variables.
- Physical laws are often not formulated in terms of event causality, but rather using differential equations $\rightarrow$ dynamic systems with cyclic (infinitesimal) causality.
- Event causality can be modeled in physics using initial conditions or external excitation functions.

Work on one of the following questions with your neighbor(s) (5–10 min)

- Can chance (randomness) be a cause?
- Try to give a definition of causality.
- Can there be final causes in science (aka teleology)?

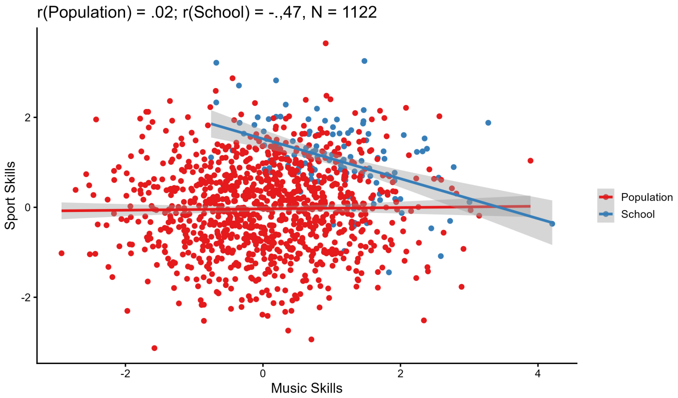
- Psychology and related fields are far removed from precise quantitative models.
- Mostly rough, qualitative formulations, undirected associations, and if-then laws.
- This type of modeling implies: no complete causal chains, many degrees of freedom, and no mechanistic models.
- Central task: to identify relevant variables (What), establish their causal relationships (Why), and quantify them (How).

How can causal relationships be determined?

- Experiments: How does Y change with respect to X, Z, U, V ...?
- In observational studies, we often have only correlations; causal relationships must often first be established.
- Two options for two variables X and Y
  - Uncorrelated: No relationship.
  - Correlated: True causality or spurious causality (or chance)
- *Example of pseudo-causality*: Inversion paradox (Simpson's Paradox, Berkson's Paradox, Lord's Paradox).

- A school admits only students who are good at either music or sports, or both.
- Assumption: In the real world, musical and athletic abilities are independent.
- Let $M \sim \mathcal{N}(0, 1)$ and $S \sim \mathcal{N}(0, 1)$ and $Cor(M, S) = 0$ in the population.
- Students are admitted to the school only if $M + S > 2$.
- A researcher visits the school and measures musical and athletic abilities. What correlation does she find?

Pseudo-causality: The Inversion Paradox



- The selection process **causes** the correlation, so this is not a true paradox.
- The conditions **at this school** are described correctly.

Correlation implies causality

But:

- The direction and nature of causality are unclear.
- The reasons are often not the ones we are actually interested in.
  - The "reason" may simply be chance.
  - The correlation may not be generalizable (e.ġ., it applies only to a sample; selection bias).
  - The reason need not lie in the "real" world, but can be purely "virtual" (e.g., mathematical operations: $cor(X, Y) = 0$, $Z = X + Y \Rightarrow cor(X, Z) > 0$.)

- The gold standard is **Randomized Controlled Trials** (RCT).
- A variable (treatment, intervention) is actively controlled or manipulated.
- *Example*: Drug trial; group A receives the new drug, group B receives a placebo; health status Y is measured.
- Random assignment of groups ensures that all other causes of an effect are randomly distributed $\Rightarrow$ they cancel each other out on average.
- The difference $E_A[Y] - E_B[Y]$ between the groups is **the average causal effect** (ACE, often also referred to as ATE).
- Additional assumptions: Everyone actually receives the same treatment; participants do not influence one another, etc.

- In experimental sciences, we always deal with random variables and probability distributions.
- **Conditional probabilities** for modeling causality: $p(Y|x)$
- Reminder: Bayes' theorem

$$p(X,Y) = P(X \cap Y) = p(X|Y)p(Y) = p(Y|X)p(X)$$
$$\Rightarrow p(X|Y) = \frac{p(Y|X)p(X)}{p(Y)}$$

- Two variables are independent if the multivariate distribution factors:

$$p(X,Y) = p(X)p(Y) \Leftrightarrow p(X|Y) = p(X), \quad p(Y|X) = p(Y)$$

- Probability theory has no concept of causality.
- In $p(X|Y)$, "given X" must be interpreted as additional information.
- Example: The statistician K. has two children.
  - *Question*: What is the probability that both children are daughters?
- K. runs into D. on the street and says: "My daughter is finally turning 18 tomorrow."
  - *Question*: What is the probability that both children are daughters?

You have 5 min to answer both question (alone or with your neighbors)

- Probability theory has no concept of causality.
- In $p(X|Y)$, "given X" must be interpreted as additional information.
- Example: The statistician K. has two children.
  - *Question*: What is the probability that both children are daughters?
  - *Answer*: We assume that there are only 2 genders $\{F, M\}$ and that the two genders are independent and equally likely, i.e., $p(\{F, F\}) = p(F) \cdot p(F) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.
- K. runs into D. on the street and says: "My daughter is finally turning 18 tomorrow."
  - *Question*: What is the probability that both children are daughters?
  - *Answer*: The four possibilities before the statement are $\{F, F\}, \{F, M\}, \{M, F\}, \{M, M\}$ with $p = \frac{1}{4}$. Given this information, we know that the case $\{M, M\}$ is impossible, so $p(\{F, F\}) = \frac{1}{3} \neq \frac{1}{4}$.

- Using the do-calculus (Judea Pearl), one can model causal relationships using probability theory.
- The do-operator "do(X)" means: we fix a value for the random variable X (intervention).
- This restricts the probability space, thereby simplifying the problem.
- This allows us to define causality: X causes Y if and only if $P(Y|do(X)) > P(Y|do(\neg X))$.
- $P(Y|do(X)) = 1$ in the deterministic case.
- With a bit of luck, the do-calculus can be used to draw causal conclusions from observational studies, and DAGs are needed for this.

Causal Networks (DAG)

- Causal networks (Directed Acyclic Graphs, DAG) encode causal knowledge through
  - **Nodes**: represent variables (e. g., A (treatment), L (covariate), Y (outcome));
  - **Arrows**: represent causal effects between variables.
- The absence of an arrow means that there is no direct causal effect.
- "Time" flows in the direction of the arrows (causes precede effects).
- There are no cycles in the graph, i.e., variables cannot influence themselves either directly or indirectly ("acyclic")

Causal Networks – Terms and Concepts

$$\begin{tikzcd} L \arrow[r] \arrow[rr, bend left] & A \arrow[r] & Y \end{tikzcd}$$

- This DAG contains all relevant three-node configurations with two arrows.
  - $L \to A \to Y$ is a **causal chain** (chain, mediation). A is the mediator of the causal effect from L to Y.
  - $L \to Y \leftarrow A$ is a **collider**. Or: Y is a collider.
  - $Y \leftarrow L \to A$ is a **confounder**. Or: L is a confounder.
- L is the **parent node** of A and Y; A is the parent node of Y.
- A and Y are **descendants** of L, and Y is a **child node** of A.
- The **skeleton** is the undirected version of a DAG.

- DAGs are interpreted as **causal Bayesian networks**; nodes are random variables, and edges indicate statistical dependence.
- do-calculus and other methods can be used to compute causal effects.
- Problem: How do you obtain the "correct" DAG?
- Expert knowledge, causal discovery, model comparison ...
- You can never be entirely certain (e.g., existence of unobserved confounders, cyclic dependencies).
- A DAG is one possible model of reality among many.
- *Example*: Is there a causal arrow between music education and school grades or intelligence?
- *Answer*: Later.

Causal Networks Exercise

Draw a DAG for the following LongGold variables. The RQ is is whether Growth or Fixed Mindset for Music is influenced by active music making (CCM). Select or combine variables if necessary. Use pen and paper, some software, or dagitty, work in groups. (5 min)

Category	Variable	Explanation
Demographic	age	Range: 11 to 16
	gender	F or M
Theory of Music	TOM.improvement	Growth Mindset ("Practising helps")
Theory of Intelligence	TOI.incremental_theory	Intelligence is changeable
Home Environment	MHE.general_score	Encouraging to make music by parents
	SES.educational_degree	Education level of parents
GoldMSI	GMS.active_engagement	Music listening and more
	GMS.musical_training	Musical education
Musical Activity	CCM	Current musical activities
Socio-emotional	GRT	Grit
	SEM.behavioral_engagement	Active engagement in school
Cognitive Abilities	MIQ.score	Visual Intelligence
	MDI.score	Melodic Memory

Download code, find `dag_exercise_vars.xls`.

- **Conditional Independence** (CI) as a central concept.

$$X \perp\!\!\!\perp Y|Z \Leftrightarrow p(X|Y,Z) = p(X|Z)$$

- X and Y are conditionally independent if, given Z, Y provides no additional information about X (and vice versa).
- *Example*: Partial correlation indicates CI for linear, Gaussian variables; χ^2 for binary variables.

d-Separation is central concept for analyzing causal pathways.

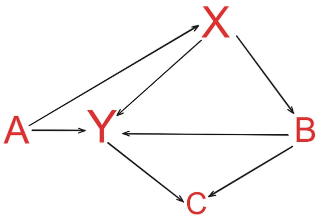
- **d-Separation**: Two nodes X and Y are d-separated if they are conditionally independent given a set of nodes Z.
- Every path between X and Y is blocked, i.e., it contains either a confounder or mediator from Z or a collider that does not originate from Z and has no descendants in Z.
- d-separation, together with the **backdoor criterion**, allows the flow of information to an outcome to be sorted and controlled.

- Backdoor criterion
  - Goal: Eliminate confounders, prevent collider bias.
  - Definition: For two nodes (X, Y), find a set of variables Z such that all causal paths are blocked (Z must not contain any child nodes of X, and Z d-separates X and Y).
  - Thus, the causal effect via do-calculus (G-estimation)

$$p(Y|do(X)) = \sum_z P(Y|X = x, Z = z)p(Z = z)$$

  - No guarantee that a Z exists ("identifiability").
  - In linear regression, one should only include variables from Z ("control for" Z) if one wants to predict Y from X.

What is a backdoor path Z for $X \to Y$?



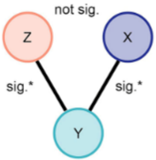
- Must not contain any colliders $\Rightarrow$ C is excluded.
- Must not contain any children of X $\Rightarrow$ B is excluded.
- Result: $Z = \{A\}$, A is a confounder of X and Y that should be controlled for; otherwise, there will be bias.

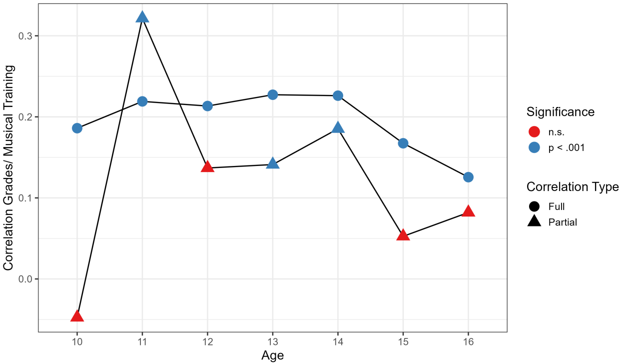
- Causal identification attempts to extract causal relationships from correlation matrices.
- Procedure:
  - Find the skeleton by systematically testing the conditional independence of node pairs and thereby eliminating edges.
  - Orient the skeleton using colliders and other rules.
- No guarantees; often only partial solutions are found (CPDAG).
- Depends on sample size and other parameters (test function, significance level, measurement errors in the data).
- Known algorithms: CI, FCI, RFCI, PC.

- Dagitty allows definition, visualization, and analysis of DAGs.
- Standalone software or by `dagitty` package for R (texttggdag for visualization).
- Functions for
  - Testing conditional independence
  - finding backdoor paths,
  - check d-separation, etc.
- *Example*. Backdoor for example graph from above

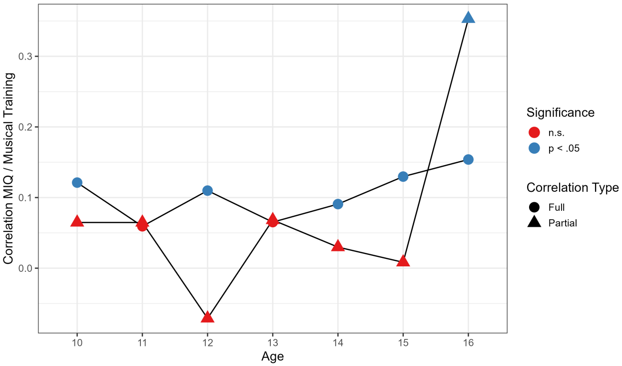
```
> dagitty('dag{
Y <- A -> X
B <- X -> Y -> C
Y <- B -> C}') |>
adjustmentSets(exposure = "X", outcome =  "Y")
> {A}
```

- In the LongGold data, school grades (AO) and IQ (MIQ) are each correlated with musical training (MT), albeit weakly ($r_{AO} \approx .2, \quad r_{MIQ} \approx .1$).
- *Question*: Is this a causal relationship, or are there one or more confounders?
- *Procedure*: For each age group, correlate overall school grades or MIQ with musical training and a third variable, once using simple correlation and once using partial correlation.
- If AO/MIQ and MT correlate marginally but not partially in the presence of a third variable, then there is a V-structure (a confounder) that may be interpreted as causal.





- For overall grades, parents' educational status (SES) is a confounder for some age groups.



- For MIQ, parents' educational status (SES) is a confounder for almost all age groups.
- **Unfortunately, music might not make you smart after all.☹**

Quantification of causal effects

- In RCTs, (almost) all "bad" causal paths are blocked by design.
- In observational studies, this is not guaranteed; that is, naive statistical methods (e. g., regression) generally yield **biased estimates of causal effects** (selection bias).
- Idea: To "artificially" turn an observational experiment into an RCT (e. g., using weighting, matching, or instrumental variables).
- With a good model, one can conduct **counterfactual experiments** (CEs) (simulations) and determine ATE and other parameters.
- Assumptions: With respect to the CE, participants behave like "**interchangeable, isolated entities without a history**"; and the CE must not contain any impossibilities ("positivity").
- Sometimes quasi-experimental data approximate RCT, but needs specific methods.

Matching / Stratification:

- For imbalanced covariates in treatment groups, find matching pairs /sets of participants with similar characteristics.
- Data pre-processing method, typically by removing cases (loss of power).
- Not unique, many different methods.
- Important methods:
  - **Propensity scores** (Rosenbaum & Rubin 1983: wildly used method: estimate probability for being treated based on covariates, e.g., using a logistic regression. Match on similar propensity scores.
  - **Coarsened Exact Matching** (CEM, Iacus, King & Porro, 2009): Coarsen variables to remove cases from cells with fewer than two cases.
  - **Mahalanobis Matching**, only for continuous variables: match on distance in Euclidean space after de-whitening covariates using a PCA.
- R package `MatchIt` offers many options and is easy to use.

Other methods

- **Inverse Probability Weighting** (IPQ): Related to matching. Uses weighted regression instead of matching/removing cases. Probabilities of being treated or not typically estimated using propensity score.
- **Doubly-robust methods** (Augmented IPW): Combining regressions with IPW for regression residuals. If at least one model is correctly specified, then the errors cancel each other out. Mitigates some shortcomings of IPW.
- **G-computation / G-Formula**. Parametric G-Formula is an extension of G-Estimation for longitudinal data with time-varying treatments. (`marginaleffects` R package does G-computation conveniently.
- **LMTP & TMLE** Longitudinal Modified Treatment Policies is a generalization of simple treatment effects, Targeted Minimum Loss Estimation is a generalization of doubly-robust estimators, (`lmtp` R package is easy to use and basically all you need.)

Other methods

- **Instrumental Variables** Method more useful for survey data, exploits quasi-experimental data by finding a proxy variables for a cause that is strongly correlated with one covariate, but not with the outcome, can serve as a quasi-treatment.
- **Difference-in-Differences** (DiD): Compares changes in outcomes over time between a treated group and an untreated control group, this eliminates confounders in a natural way by differencing.
- **Regression Discontinuity Design** (RDD): Uses a sharp numerical threshold or cutoff to determine who receives treatment, e.g., by policy changes or differences, or selection mechanism such as entrance exams or scholarships by merit. (Still subject to confounding.)

	IPW	G-computation	TMLE	SDR
Simple to implement	$\star$			
Uses outcome regression		$\star$	$\star$	$\star$
Uses the propensity score	$\star$		$\star$	$\star$
Valid inference with machine learning			$\star$	$\star$
Substitution estimator		$\star$	$\star$	
$\tau + 1$ doubly robust			$\star$	$\star$
Sequentially doubly robust				$\star$
Recommendation	Don't use	Don't use	**Recommended**	Use (only dyn.)

Cons G-computation, IPW: Correctness of bootstrap requires pre-specified parametric models, consistency requires correct estimation of all regressions.
Sequential and doubly robust can be worse if both propensity and outcome models are misspecified (Kang & Schafer, 2007)!

- Simulated simple data set
- Exercise
- Flying Steps data (maybe)

Exercise: Misspecification and Measurement Error

- Go to the following link and download simulated data
- Try estimating ATE using
    - `a` instead `t` (treatment with error)
    - Additional covariate `z` (collider)
    - `z` instead of `l1`
    - `l1_me` instead of `l1` (l1 with measurement error)
- with `MatchIt` or `lmpt`.

Finally, the End

- Causal inference attempts to solve two problems
  1. Identification of causal relationships based on correlations (and domain knowledge).
  2. Unbiased estimation of causal effects (ATE).
- Causal DAGs are the central tool.
- These methods have seen little application in (music) psychology so far, but could be helpful.
- DAGs are generally helpful for gaining clarity about the relationships and variable types in the data, especially when there are many variables. ($\Rightarrow$ "Improve your regressions.")
- There is a close connection to Structural Equation Modeling (SEM).
- You should clarify the DAG **before data collection** to ensure that you measure only the truly relevant variables.

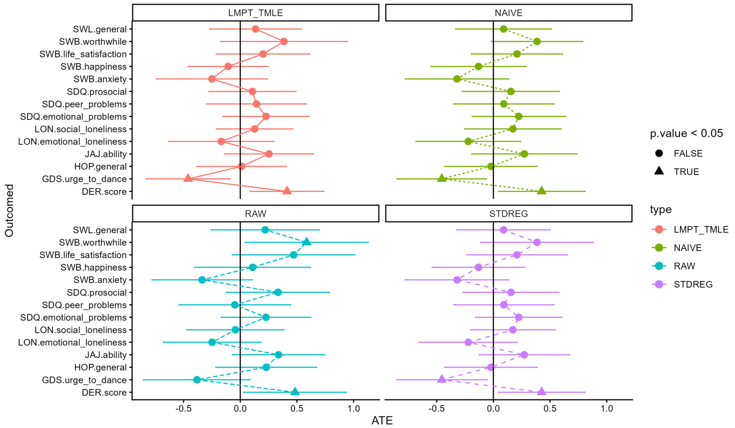
Thank you! Questions?

- Flying Steps Education provides (street/urban) dance classes for Berlin schools since 2024 (in part substituting for regular PE lessons).
- In 2025, we had the occasion to run a pilot study to assess effects on dance classes on well-being, social emotional development and cognitive abilities (Visual Working Memory).
- Two measurement time points ($\approx$ 50 d apart) for kids with and without dance lessons.
- Battery of established measures, demographics, Big5, physical activities, dance-related questions.

Example TMLE: Flying Steps

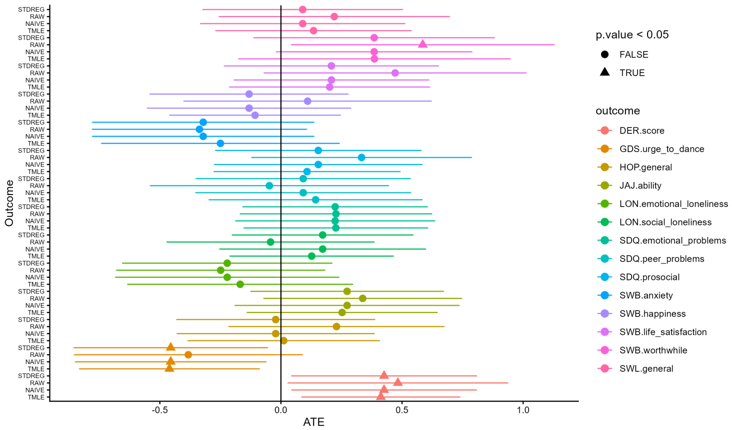
- We compared different ATE estimation methods using demographics, Big5, and physical activities (PAC) as covariates
- 14 outcomes: SDQ (school difficulties, 3 subscales), LON (Loneliness scale, 2 subscales), HOP (Hope, 1 subscale), SWB (subjective well-being, 4 subcales), SWL (Worth living, 1 subscale), JAJ (Visual Working Memory), GDS (subscale urge to dance), DER (Dance Emotion Recognition test).
- ATE for baseline, as kids had already dance lessons.
  - Raw data differences (RAW)
  - simple regression estimates (NAIVE)
  - standardization with the `stdReg` package (STDREG)
  - TMLE with the `lmtp` package (TMLE)
  - N = 94, N_{control} = 28., N_{dance} = 68
- ATE for both time points (static treatment), only TMLE.

Flying Steps: ATE at baseline for 14 outcomes



Krippendorff's α: Estimates: .9, Significance: .71

Flying Steps: ATE at baseline for 14 outcomes



Flying Steps: Full ATE for 14 outcomes